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Mixture of Limits: Navigation Law for Computer Intelligence

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Measurement. Research study, not a journal final. Energy figures from the software reference are OpCounter analytical estimates, not board power. We have not synthesized or metered an FPGA board. No fabricated citations.

Floors where more bits stop buying outcomes. Compression and predictive formulas beat excess enumeration; neural nets are a demoted residual leaf. Soft-ref constructive existence in software; not FPGA or board package energy.

Abstract

Mixture of Limits is a navigation law for computer intelligence: there exist **floors** past which additional tokens, parameters, or joules do not purchase verifiable progress on a task coordinate. The law is information-theoretic and rooted in physics. Across human scientific history, compression into predictive formulas and invariants has repeatedly beaten excess enumeration of observations: from Kepler's laws over Tycho's tables, through Newton's closed forms, to Shannon's bit accounting and Landauer's thermodynamic floor on irreversible erasure. Mixture of Limits operationalizes that lineage for machines: **Lookup** → **Formula** → **Solver/settle** → **Model LAST**, with close owned as propose → certify → commit/refuse → receipt.

The industry default escalates Mixture-of-Experts (MoE) capacity *inside* a generative corridor. Mixture of Limits instead names floors *outside* generation: Value of Information (VoI), grammar coverage, Landauer/joule estimate, certificate, and settle-refuse. It demotes the neural net to a residual leaf. Soft-ref mol prove is constructive existence in software. Catalog and Landauer figures on receipts are **estimates**; package measured_j appears only when a labeled meter returns a reading.

1. Introduction: pursuit of limits toward AGI

The dominant construction for computer intelligence (CI) treats the neural net as the substrate and excess tokens as the growth variable. That construction diverges from the historical method that produced reliable science. When Brahe enumerated planetary positions, the tables were necessary

but not the law. Kepler compressed years of positions into three predictive relations. Newton further compressed those relations into invariants and closed forms that travel across domains. The same pattern recurs wherever a grammar of nature is covered: a short formula outruns an infinite ledger of cases.

Information theory names the modern accounting of that pattern. Shannon (1948) prices bits under uncertainty as settled law. Value of Information (Vol) prices whether another observation is worth its cost for a decision. Landauer (1961) prices irreversible bit erasure in joules at temperature (T): $(E_{\text{bit}} = k_B T)$ per bit erased in the ideal model. That Landauer quantity is a thermodynamic lower bound and, on Mixture of Limits receipts, a **labeled estimate**—not a wattmeter reading. Together they imply floors: past a point, more bits stop buying outcomes that matter for a stated benefit.

Thesis. Pursuit of limits is the path toward AGI-grade reliability. Excess-token and MoE scaling diverge from true Vol for a given benefit. Mixture of Limits is the executable navigation law that binds those floors (Vol, grammar, Landauer/joules as estimate, certificate, settle-refuse) against industry Mixture-of-Experts. **Addendum:** Shannon, Landauer, Kolmogorov/Solomonoff/Chaitin, Howard (Vol), and formula-first science AI already state these floors as settled theory. The remaining bottleneck is **applied mathematics embodied in materials and physical hardware**. **Frontier addendum (§8.4-§8.12):** deeper latents, energy-based hybrids, world models, test-time compute, super learning, architecture fads (including Jamba-class hybrids and mixture of experts), and algorithmic efficiency wins still improve generators or ensembles; they do not cancel Vol, grammar, energy, or certify floors. Mixture of Limits remains the navigation law, not a horse race among generators. The hardware-economics reading (§8.12) states that computers are hardware, software is applied engineering under constraints, and floors bind even when optimized algorithms take market share on cheaper devices.

This paper's contribution is the study prose for that law as published on research.openie.dev, grounded in the clean-room mixture-of-limits software reference (Apache-2.0 OR MIT). Companion studies already on this site supply the commit-record interface ([Notational Intelligence as Commit Law](#)) and the economic stop after free digital inference ([Satiation and Scarcity after Free AI](#)). Mixture of Limits is the navigation law those companions sit under: NI owns irreversible commit shape; Satiation owns economic done; Mixture of Limits owns *which gear closes* and *when refuse is success*.

Three measurement facts constrain every later number. First, joules from catalog surrogates and OpCounter-style analytics are **estimates**, not board power. Second, Landauer annotations are **estimates**, distinct from RAPL/NVML/measured_j. Third, board_synth_claimed=false; package measured_j is set only when a Metered probe returns a reading.

Scope. Section 2 states the law and Periodic Stack navigation. Section 3 states the proof spine $(E(x)(D)(S,V))$. Section 4 describes the cascade and close/receipt bind. Section 5 bridges to Satiation without rewriting it. Section 6 maps prove[?]claim. Section 7 sketches Vol, grammar, and settle-refuse mathematics. Section 8 places related work as a verified citation chain (historical→recent proof points) plus Tier A formula/mechanism systems, then teaches latent space, Logical Intelligence (energy-based model / large language model / latent hybrid), World Labs (spatial world models), test-

time inference (test-time compute), super learning, a survey of live 2025–2026 approaches, the AI21 Jamba hybrid story versus later DeepSeek / GLM efficiency moves, and the hardware-economics thesis, with the addendum that even projected-superior methods still hit floors. Section 9 contrasts Mixture of Limits with Mixture-of-Experts and situates model-to-model field systems. Section 10 opens toward AGI via limits.

2. Law: floors where more bits stop buying outcomes

Mixture of Limits asserts: for a task coordinate, there exist floors such that additional tokens, parameters, or joules do not purchase verifiable progress. Intelligence is therefore a **navigation law** over a structured stack of primitives; not an invitation to dump residual capacity into a generative corridor.

Consequences:

1. **Mathematical compression / formulas beat excess tokens** when the grammar is covered.
2. **Neural nets are a demoted residual leaf**, not the default substrate of CI.
3. Routing is **cheapest-sufficient**: escalate only on miss; refuse when escalation is unsafe or Vol-negative.
4. Proof / energy law: $(E(x) (D)(S,V))$ with labeled joule receipts; Landauer labeled as estimate.
5. **Available devices / multi-fabric**: after a cascade gear closes, pick the cheapest sufficient device class; not one accelerator by default.

Named floors

Limit id	Kind	Meaning
voi	ValueOfInformation	Marginal bits worthless under budget. stop
grammar	GrammarCoverage	Covered by Lookup/Formula/Solver; or refuse unknown
energy	Energy	Joule budget floor; Landauer-aware estimates
efa_certificate	Certificate	Diverge / fail certify → refuse before commit
settle_refuse	SettleRefuse	Ternary / energy-landscape settle will not settle → refuse
primitive_gap	PrimitiveGap	Missing Periodic Stack primitive; surface the gap
safety / wca_refuse	Safety / WCA	Capability deny; WCA certify refuse
information / latency	Information / Latency	Sufficiency and soft wall-clock ceilings

Neural net demotion

The model tier is **LAST**. Default budgets set allow_model=false. Even when allowed, the model remains a leaf: it may propose residual content; it never bypasses certify-before-commit, and ModelGenerated answers cannot be laundered as Deterministic. Coin-cell / edge power is O(1) formula plus LUT refuse, not a tiny transformer.

Periodic Stack

Mixture of Limits navigates the OpenIE **Periodic Stack** (compute.openie.dev: **258 primitives / 33 families**). The soft-ref proves an in-tree **subset** navigator with Present/Gap cells and scale notes citing 258/33. Full live catalog and μ calibration corpora remain outside soft-ref proof scope. Empty cells surface as primitive_gap.

Mixture of Limits is not MoE

MoE (generative corridor)	Mixture of Limits (navigation law)
Experts live <i>inside</i> a model; gate selects parameters	Limits live <i>outside</i> generation; floors stop escalation
More experts → more capacity for the same generative act	More named floors → sharper refuse / cheaper close
Success = next-token likelihood	Success = closed grammar + certified commit + joule receipt
Model is the substrate	Model is a demoted residual leaf

Clean-room policy: Mixture of Limits is law + runtime, not a wrapper on MoE, and not a path-dependent fork of sibling OpenIE trees for the proven path.

3. Proof spine: (E(x) (D)(S,V))

The energy / impedance law used throughout Mixture of Limits is:

[E(x) (D)(S,V)]

where (E(x)) is path energy for closing act (x), ((D)) scales with decision / difficulty structure (D), and ((S,V)) is an impedance factor for stack coordinate (S) and view (V). Soft-ref receipts stamp catalog () with mu_source=catalog and a landauer_floor_ratio. The product is an **analytical / catalog estimate**, not a package joule reading.

Landauer (estimate only). For irreversible erasure of (n) bits at temperature (T),

[E_{(n,T)} = n,k_B T]

Receipts may annotate `landauer_floor_J` from this formula. That annotation is **estimate ≠ measured**. Soft-ref closes keep `measured_j=None`. Optional OS meter features (RAPL / IOReport / powermetrics), when present and successful, may populate `measured_j` only under a labeled `MeasureSource`; failure or VM stays unavailable. Soft-ref prove criteria do not require those meters.

Historical rhyme. Kepler did not need every future observation once the three laws closed the grammar of planetary motion for the epoch. Newton did not need a larger ephemeris table to predict a new orbit once ($F=ma$) and inverse-square gravitation covered the coordinate. Shannon did not need infinite samples to bound channel capacity. Landauer did not need a particular chip to state a thermodynamic lower bound. Mixture of Limits spine is the same move for CI: bind a floor, close cheapest-sufficient, refuse when the floor says stop.

Constructive existence (software). Soft-ref `cargo run -p mol-cli -- prove` prints VERIFIED per criterion and exits 0 (~29 soft-ref criteria spanning deterministic close, formula/lookup without model, Vol refuse, settle commit+refuse, certificate refuse, capability default-deny, receipt labels, replay-class coercion deny, Periodic Stack subset navigation, μ catalog, transcript replay, Z2 cite / Z1 compose, agent mailbox, bitemporal memory, fabric routing, desktop headless shell, OS meter labels, WASM capsule, Agent Lane, multi-fabric receipts, ecosystem e2e certify; see PLAN.md in the Mixture of Limits workspace). That is constructive existence **in software**—a runtime fact, not a board energy measurement. Prove[?]claim mapping is Leapfrog-owned (§6).

4. Cascade and close: Lookup → Formula → Solver → Model LAST

MathGround / OpenIE cascade order (type-enforced replay classes):

Lookup → Formula (closed-form) → Solver / settle → Model LAST

Mixture of Limits tier	Role	Replay class (typical)
Lookup	LUT / allow-table / registry hit	Deterministic
Formula	Closed identity / unit convert / law instance	Deterministic
Solver	Sparse solve / ternary settle	Deterministic or compose under Solver
Model	Stochastic residual only	ModelGenerated

Invariant: `ModelGenerated` →/Deterministic. Typed answers refuse coercion.

Automation / close loop:

sense → classify grammar → cheapest sufficient → certify → commit|refuse → receipt

Mixture of Limits **owns close**. Refuse over escalate when unsafe or Vol-negative. Refuse is a **lawful success** and still emits a receipt.

co/0 bind sketch (namespaces)

Soft-ref receipt labels bind act namespaces in the spirit of OpenIE co/0 receipts (sketch, not a live leapfrog path-dep):

Namespace	Role
claim.*	Seeded / retrieved knowledge claims (Z2); unknown → refuse, no invent
act.*	Capability-gated acts; Mutate/External default-deny
close.*	Route + certify gate; commit or refuse
receipt.*	estimated_j labeled; measured_j only if Metered probe returned; board_synth_claimed=false

Rules stamped on every soft-ref close:

1. **Certify-before-commit**; no irreversible side effect without certificate pass.
2. **ModelGenerated** → **Deterministic**; sealed replay markers.
3. **measured_j only if Metered**; otherwise None / unavailable.
4. **Refuse = lawful success + receipt**. Vol / settle_refuse / certificate / capability denies still close the ledger.

Multi-fabric soft-ref: after tier selection, route cheapest sufficient DeviceKind (Cpu always present; Gpu*/Wasm/ThermoSettle/... optional). Detection ≠ joules.

5. Econ bridge: Satiation (cite, do not rewrite)

Digital inference prices can fall toward free-at-margin without implying free energy, free actuation, or unbounded value after a chore is complete. The companion study [Satiation and Scarcity after Free AI](#) defines satiation as a stop on a written completeness predicate and cites published inference-price series. Mixture of Limits supplies the *machine* stop that matches that economic intuition: Vol and grammar floors refuse excess generation once benefit is covered; commit/refuse receipts (NI) record the irreversible boundary.

Satiation owns economic done and its price tables; NI owns the commit record; Mixture of Limits owns navigation floors that make “done” and “unsafe” executable without opening the model leaf.

6. Prove claim map

The Mixture of Limits constructive prove is not a leaderboard stunt. It is an **information-theoretic** close law rooted in **physics**: the same historical arc that took astronomy from excess epicycles to Kepler/Newton predictive floors, and that takes computation from Shannon → Landauer → complexity / Vol toward named energy floors. Global academia already holds those proof points; the bottleneck is **applied math × materials (hardware) embodiment**, not awareness of the slogans. This section only maps what mol prove actually stamps VERIFIED onto paper claim.* ids: embodiment of known floors, not invention of new physics.

A claim is **proven** only when the in-tree harness prints VERIFIED for the named criterion and exits 0. A claim is **soft-ref only** when PLAN §6 marks it OUT OF PROOF SCOPE, or when the criterion itself stamps soft-ref / fixture / feature-gated behavior rather than live silicon. Proven-path rows keep measured_j=None (or Metered only when a probe returns) and board_synth_claimed=false.

Measurement labels used below (OpenIE measurement grammar):

Label	Meaning
Unmetered	Software-ref / soft-ref path; measured_j=None; estimate may be labeled
Estimated	Catalog / Landauer / fuel / μ surrogate with explicit EstimateKind or mu_source=catalog
Metered	Optional live probe (energy-meter) returned numbers (not required for prove)

Default close path for all proven claims: **Unmetered** (+ **Estimated** where μ / Landauer / fuel appear). board_synth_claimed=false everywhere in the proven path.

6.1 Harness meta

Prove #	Criterion (PLAN)	Paper claim id	Status	Measurement
P0	cargo test --workspace green	claim.mol.ci_workspace	Proven (CI / operator; not printed by mol prove)	-
P1	cargo run -p mol-cli --prove exits 0	claim.mol.prove_harness	Proven (prints VERIFIED per row below)	Unmetered

6.2 Proven claims (P2-P24)

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
P2	Deterministic close (double-run)	<code>claim.mol.deterministic_close</code>	Same asks yield the same commit/refuse + limit id.	-	Un
P3	Formula/lookup close without model	<code>claim.mol.formula_lookup_no_model</code>	Landauer / unit-convert closes COMMIT at Lookup or Formula; model never answered.	-	Un Est (La floc rec
P4	Vol refuse when !allow_model	<code>claim.mol.voi_refuse</code>	Free-form ask → REFUSE voi.	-	Un
P5	Settle commit + settle refuse	<code>claim.mol.settle_ternary</code>	Ternary settle → COMMIT; will not settle → REFUSE settle_refuse.	Software-ref settle stub; no klere-vm / FPGA	Un
P6	Certificate refuse on diverge	<code>claim.mol.efa_certificate_refuse</code>	Formula + diverge → REFUSE efa_certificate.	In-tree EFA-style cert; Ferric / live BMI loop soft-ref	Un
P7	Capability default-deny mutate	<code>claim.mol.capability_deny_mutate</code>	<code>AutomateGate::default().gate(Mutate)</code> → Refuse.	-	Un
P8	Receipt labels	<code>claim.mol.receipt_honesty</code>	Software-ref: <code>measured_j=None</code> , <code>board_synth_claimed=false</code> , estimate labeled (<code>EstimateKind</code>).	Live RAPL/ NVML default close OUT OF SCOPE	Un Est
P9	ModelGenerated → Deterministic	<code>claim.mol.replay_no_strengthen</code>	<code>TypedAnswer::weaken_to</code> returns <code>ReplayCoercion</code> .	-	-
P10	Periodic Stack subset navigation	<code>claim.mol.stack_subset_nav</code>	Family + present primitive + scale → COMMIT at Lookup; note cites 258/33.	Full 258 live catalog	Un

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
				OUT OF SCOPE	
P11	primitive_gap via registry probe	claim.mol.primitive_gap	Gap marker physical_settle + absent name → REFUSE primitive_gap (not string-only).	-	Un
P12	μ / impedance catalog	claim.mol.mu_catalog	Receipts stamp mu_source=catalog, mu, landauer_floor_ratio; $E \approx \theta \cdot \mu$ (catalog estimate, not RAPL/measured_j).	μ calib corpus OUT OF SCOPE	Est
P13	Receipt transcript replay	claim.mol.receipt_replay	JSONL/in-memory replay reproduces commit/refuse + limit id; model never answered.	-	Un
P14	Z2 retrieve+cite	claim.mol.z2_retrieve_cite	Seeded factual hit → COMMIT RetrievedCited + citation ids; unknown → REFUSE claim_unknown (no invent).	Seed corpus in-tree only	Un
P15	Z1 compose/ synthesis	claim.mol.z1_compose	≥ 2 cited claims → COMMIT Composed + synthesis.composed_from; missing → REFUSE compose_missing; never laundered as Deterministic/RetrievedCited alone.	-	Un
P16	Agent mailbox loop	claim.mol.agent_mailbox	Goal/Message/Act → close → commitl refuse + receipt in transcript; demo hits formula/cite/compose COMMIT and voi/ settle_refuse REFUSE; model cold; measurement flags hold.	-	Un
P17	Bitemporal state + memory	claim.mol.bitemporal_memory	Writes only on Mixture of Limits close COMMIT; Mutate deny / free-form remember → REFUSE; recall → RetrievedCited + memory: cite; unknown → memory_unknown.	-	Un
P18	Fabric routing + soft-ref inventory	claim.mol.fabric_soft_ref	Offline inventory valid (source=software_ref, Cpu always); Lookup/Formula→Cpu; Settle→ThermoSettlelCpu; Model→Gpu* or REFUSE fabric_unavailable; detect $\neq$ measured joules.	Live wgpu presence optional / not required for prove	Un
P19	Desktop shell headless	claim.mol.desktop_headless	ShellSession ask/close → receipt view (zone/fabric/limit/estimated_j/	egui GUI optional	Un Est

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
			measured_j=None); joule ledger; Vol refuse; GUI not required.		
P20	OS meter labels	claim.mol.os_meter_honesty	Feature off / VM / no permission → measured_j=None, empty components, measure_source=unavailable; fixtures stamp labeled sources; default close stays unmetered.	Live probe optional (energy-meter)	Unr (de Me whe retu
P21	WASM capsule certify	claim.mol.wasm_capsule	add.wasm under stub runtime; fuel → estimated_j only; FS/net grant / fuel exceed / host-share → REFUSE; close stamps capsule.	Optional wasmtime feature	Unr Est (fue
P22	Agent Lane session path	claim.mol.agent_lane	Lane partitions; host_invoke without provenance refuses; keyword confirm insufficient; need GrantReceipt; allow stamps AgentLaneReceipt.	-	Unr
P23	Multi-fabric compute receipt	claim.mol.fabric_compute_receipt	Soft-ref inventory; ComputeFabric / ScheduleDecision / ComputeStepReceipt with fabric_id; cascade stamps fabric:route; CPU Formula COMMIT; Model residual may fabric_unavailable; measured_j=None on soft-ref path.	Ferric not path-dep'd	Unr
P23b	Live fabric soft	claim.mol.fabric_live_soft	Feature off stays software_ref; mock/live Metal can stamp gpu_metal; Formula stays Cpu; missing Gpu* fail-closed; prove does not require a GPU; measured_j=None unless Metered.	Soft / optional detect	Unr
P23c	wgpu / Metal tiny kernel	claim.mol.kernel_vector_add	Clean-room WGSL vector-add; soft stub offline (same checksum); Gpu* stamps kernel:vector_add + execution_proof; optional overlapping meter may stamp package measured_j with energy_honesty=measured (field name); no rail-sum package joules.	Live Metal optional; fixture OK for prove	Unr (de Me if m retu
P24	Ecosystem e2e certify (one receipt)	claim.mol.ecosystem_e2e	Lane → fabric (Cpu soft-ref) → WASM → energy labels (estimated fuel; optional meter_sample) → host invoke only with GrantReceipt → commitlrefuse on one receipt; refuse-without-grant → grant_receipt_required.	-	Unr Est

6.3 Soft-ref only (not proven; do not escalate to paper results)

These are PLAN §6 residuals. Citing them is allowed as roadmap / reference semantics. Treating them as measured results is not.

Soft-ref item	Why soft-ref	Related prove row (if any)
Live silicon RAPL / NVML as default close	Optional energy-meter; soft-ref prove stays measured_j=None	P8, P20
Live wgpu adapter presence	Feature fabric-detect; inventory soft-ref remains prove default	P18, P23b, P23c
Ferric / on-device EFA (BMI hardware loop)	Reference semantics only; not path-dep'd	P6, P18, P23*
klere-vm WASM / FPGA meter (real pJ)	Software-ref StubKlereSettle	P5
Live openie-path / leapfrog ask bridge	Adapter port	-
Live WCA MCP / wca-lut-edge in-proc certify	Adapter port	-
Full Periodic Stack 258 live catalog + μ calib corpus	Subset navigator + Gap + tier μ catalog are in proof	P10, P12
Trained weights / candle / tract Model leaf	Model stays demoted stub	P3, P4, P16

6.4 Seeded knowledge claims (Z2 corpus; not the prove map)

In-tree ClaimStore seeds used by P14/P15 (retrieve/compose), distinct from the paper claim ids above:

- claim:landauer.principle
- claim:mol.law
- claim:mol.proof_law
- claim:mol.cascade
- claim:stack.scale
- claim:honesty.no_fake_rapl
- claim:replay.retrieved_cited
- claim:openie.z2

6.5 Measurement scope of the proven map

- Soft-ref prove path: measured_j=None (Unmetered); Metered only when a live probe returns.
- board_synth_claimed=false on the proven path.
- Soft-ref v0.1 prove is software constructive existence, not board / FPGA watt leadership.
- Soft-ref roadmap items (§6.3) stay roadmap; they are not escalated to proven claims.

7. Math sketches: Vol stop, grammar coverage, settle-refuse

7.1 Vol stop

Let (B) be a benefit functional for a decision, (x) the current information state, and (c) the cost (bits, latency, or estimated joules) of acquiring observation (y). A simple stop rule:

$$[(x,y) = [B(x,y) - [B(x)] - c(y)]$$

Refuse escalation when ((x,y)) (or () for a configured threshold). Soft-ref Mixture of Limits encodes this as floor voi with allow_model=false by default: free-form asks that would only open the model leaf refuse rather than spend. This is the Shannon/decision-theoretic cousin of Kepler stopping new naked-eye points once the law predicts within tolerance. Domain-calibrated Bayesian Vol tables remain a deployment parameter, not a soft-ref prove deliverable.

7.2 Grammar coverage

Let (G) be the covered grammar (Lookup registry U Formula identities U Solver settle programs). A query (q) is **covered** if a deterministic derivation (q _G a) exists with replay class not ModelGenerated. Cascade order tries Lookup, then Formula, then Solver. If covered, **do not open the model**. If uncovered and policy forbids model escalation, refuse (grammar / voi / primitive_gap as applicable). Historical analog: once Newton's grammar covers orbital mechanics for the force law at hand, one does not re-enumerate Brahe's table to answer a new initial-condition query.

7.3 Settle-refuse

Under Solver, a ternary / energy-landscape settle either reaches a certified fixed point (commit) or signals will-not-settle (refuse settle_refuse). Soft-ref implements this as real in-tree settle logic plus certificate diverge tags (efa_certificate). Thermodynamic-settle inspiration is conceptual: soft-ref v0.1 has no annealer drivers and no adiabatic wall joules. Landauer appears as a labeled estimate on receipts.

7.4 Bound sketch tying Shannon and Landauer

For (n) irreversible erasures on a path that closes a decision,

$$[E_{\{}} E_{\{}(n,T) = n,k_B T]$$

and the Mixture of Limits catalog estimate further requires (E_{\{}(x) (D)(S,V)). measured_j requires a Metered probe. Excess tokens that do not change (B) fail the Vol test even when joule budgets remain; the information floor binds first.

8. Related work: proof points, lineage, and Tier A

8.0 Academic proof points: top minds already named the floors

Mixture of Limits is an executable navigation law over floors that information theorists, physicists, and decision theorists already stated as settled theory, and that formula-first science AI keeps rediscovering in modern form. Below is a **historical** → **recent citation chain** with DOIs verified for this draft (or marked TBD). These are global academic proof points that compression, floors, and formulas beat excess enumeration.

Information, complexity, and physical cost

Proof point	Who / when	What they already said	Cite (verified)
Bits under uncertainty	Shannon (1948)	Quantitative communication limits; entropy as missing information	10.1002/j.1538-7305.1948.tb01338.x (Part I); Part II 10.1002/j.1538-7305.1948.tb00917.x
Irreversible erasure costs heat	Landauer (1961)	Logical irreversibility $\Rightarrow$ minimal heat $\sim(kT)$ per irreversible function	10.1147/rd.53.0183
Reversible computing	Bennett (1973)	Computation can be logically reversible; Landauer cost is about erasure, not “computing” per se	10.1147/rd.176.0525
Algorithmic complexity	Kolmogorov (1965/1968)	Quantity of information as shortest program length; compression as definition	Russian original <i>Probl. Peredachi Inf.</i> 1(1):3–11 (1965), no DOI; English transl. 10.1080/00207166808803030 ; IEEE note 10.1109/TIT.1968.1054210
Algorithmic probability / induction	Solomonoff (1964)	Prefer short programs; universal prior over computable regularities	Part I 10.1016/S0019-9958(64)90223-2
Algorithmic incompleteness	Chaitin (1977)	Algorithmic information bounds what finitely axiomatized theories can prove	10.1147/rd.214.0350
Probability as extended logic	Jaynes (2003 book; earlier papers)	Maximum-entropy / logic-of-science discipline against ad-hoc excess modeling	10.1017/CBO9780511790423
	Cover & Thomas		

Proof point	Who / when	What they already said	Cite (verified)
Textbook IT + decision links		Canonical Elements of Information Theory (2nd ed.)	Wiley book DOI 10.1002/047174882X
Value of Information	Howard (1966)	Shannon bits $\neq$ decision value; Vol joins probability with economic consequence; stop when information is worthless for the decision	10.1109/TSSC.1966.300074
Physics of computation lectures	Feynman (pub. Hey/Allen eds.)	Physical limits of computing, reversible/thermo themes in lecture form	Anniversary ed. 10.1201/9781003358817 ; 1996 Addison-Wesley ISBN 0-201-48991-0

Historical mathematical physics (compression beats enumeration). See §8.1:

Brahe→Kepler→Newton is the pre-IT proof that tables yield to laws. Imprint DOIs for critical editions of *Principia* remain **TBD** if a journal version requires them.

Recent formula / mechanism / certify systems (Tier A inventory), from docs/adjacent-field-hunt.md and docs/appendix-mwm-ai-newton.md (2026-09-30):

Proof point	Who	Mixture of Limits reading	Cite (verified)
AI Feynman	Udrescu & Tegmark	NN decomposes; closed form is the commit	10.1126/sciadv.aay2631
SINDy	Brunton / Proctor / Kutz	Sparse grammar; reject dense residual fits	10.1073/pnas.1517384113
PySR	Cranmer	Production symbolic regression	arXiv: 2305.01582 (journal DOI TBD if distinct)
AI-Newton	Fang et al.	Concept library + general laws across experiments	arXiv: 2504.01538
AlphaGeometry / AlphaProof	DeepMind	Propose → symbolic/Lean certify (hard floor)	Nature 10.1038/s41586-025-09833-y
Mechanistic World Models	Posner / Lei / Schölkopf	Mechanisms > predictive MoE modules	arXiv: 2607.12474
AutoSINDy hybrid	(hunt 2026 note)	PySR → library → SINDy	arXiv: 2605.09696

Raiffa-style decision analysis texts are the pedagogical cousins of Howard's Vol; a specific Raiffa imprint DOI is **TBD**.

8.0.1 The gap is embodiment: not awareness

Claim (addendum). Applied mathematics × materials science (physical hardware) is the bottleneck, **not** awareness of what should be done.

Evidence for the awareness side is the table above: Shannon priced bits; Howard priced Vol for decisions; Landauer/Bennett priced irreversible erasure; Kolmogorov/Solomonoff/Chaitin priced shortest programs and incompleteness; Feynman taught physical limits of computation; SINDy/PySR/AI Feynman/AI-Newton/AlphaGeometry/MWMs keep showing that formulas, sparsity, verification, and mechanisms beat unbounded enumeration and unguided MoE-style prediction.

What remains hard, and what Mixture of Limits companions (NI commit law; coin-cell Formula+LUT; multi-fabric routing; Metered vs Estimated labels) point at, is **embodying** those floors in real stacks: materials and devices that make Lookup/Formula/Solver cheap, certificates enforceable before irreversible acts, and joule receipts that keep estimates separate from measured_j. Soft-ref mol prove shows the *law+runtime* exists in software. Materials and device embodiment remain open. Soft-ref path: board_synth_claimed=false; package energy only when Metered.

8.1 Historical lineage: compression beats enumeration

Tycho Brahe → Kepler. Brahe's program was high-fidelity enumeration. Kepler's three laws compressed those tables into predictive relations. Enumeration was necessary input; the law was the compressible object that stopped needing every future observation.

Kepler → Newton. Newton folded planetary regularities into invariants and closed forms ($F=ma$; universal gravitation). Predictive power migrated from case tables to formulas that travel. AI-Newton (Fang et al., arXiv:2504.01538) is a contemporary soft echo: concept libraries and general laws across many noisy mechanics experiments. NN as recommender residual, not substrate.

Invariants and closed forms. Symmetry → conservation (Noether tradition), Hamiltonian/Lagrangian reduction, and integrals of motion are the mathematical-physics statement of "formula wins when grammar is covered." The Mixture of Limits Formula gear is that tradition's CI leaf.

Information theory and physical cost. Shannon, Howard (Vol), Landauer, Bennett, Kolmogorov/Solomonoff/Chaitin, and Cover & Thomas form the IT/decision/thermodynamic chain in §8.0. Mixture of Limits binds them as named floors and labeled estimates. Soft-ref Landauer stamps are lower-bound estimates, not claims of near-bound silicon operation.

Notational intelligence. Lee's notational-intelligence claim is the companion frame for executable predicates; see [NI Commit Law](#).

8.2 Tier A detail: formula / mechanism / executable physics

Inventory dated 2026-09-30 in the Mixture of Limits workspace hunt docs. Capsule readings:

1. **AI Feynman (+ 2.0):** Physics-inspired symbolic regression; NN for symmetries/separability, then recursive formula recovery. Canonical *Formula* gear.

2. **SINDy**: Sparse identification of nonlinear dynamics; explicit grammar + sparsity floor.
3. **PySR**: Evolutionary symbolic regression; distill NNs → closed forms.
4. **AI-Newton**: Concept-driven law discovery across noisy mechanics experiments.
5. **AlphaGeometry / AlphaProof**: LM proposes; verifier is the hard floor (propose → certify → commit/refuse template).
6. **Mechanistic World Models**: Sister philosophy: reusable mechanisms; MoE as wrong module semantics for explanation.

Additional hunt rows (AtomAgents/SciAgents, KeplerAgent/NewtonBench, AI-Descartes, DreamCoder/COMET/NEO, Robot Scientist) sit as hybrid or mechanism-library cousins. Where a DOI was not verified beyond arXiv IDs, cite arXiv and mark journal DOI **TBD**.

Mixture of Limits reading of Tier A. Formula/mechanism discovery (A) plus physics-executable leaves (B in the hunt) beat token-escalation workflows (C). MoE-style mixtures are the wrong “Mixture of.”

8.3 What Mixture of Limits adds

Tier A systems recover or verify formulas. Classical IT already priced bits, Vol, and erasure. Mixture of Limits names the **navigation law** that decides *when* Lookup/Formula/Solver suffice, *when* to refuse, and *how* receipts label estimated vs measured joules, then implements a clean-room soft-ref close loop. Soft-ref constructive existence is software constructive existence under those floors—not a materials breakthrough or a new physics corpus.

8.4 Latent space: depth buys compression, not freedom from floors

A **latent space** is a compressed coordinate system for states that matter to a task: continuous vectors (dense embeddings), discrete codes (VQ / codebook indices), or hybrids. Depth in a latent buys what Shannon and Kolmogorov already price: a shorter description of structure that would otherwise be enumerated as raw observations, pixels, tokens, or traces. Autoencoders, VAEs, diffusion latents, and continuous reasoning traces are modern instances of that move. The representation can make Lookup hits cheaper, Formula edits local, and Solver landscapes smoother.

Representation is not law. A deeper latent does not erase the floors Mixture of Limits names:

1. **Value of Information (Vol)**. Extra latent dimensions, longer continuous traces, or denser codebooks are more bits under a budget. When marginal latent bits do not change the decision benefit (B), Vol says stop. Excess latent enumeration is the same failure mode as excess token enumeration, only in a different alphabet.
2. **Grammar coverage**. If Lookup, Formula, or Solver already covers the coordinate, opening a latent generator is waste. Covered grammar closes before Model. A latent that rediscovers a closed form is still Model residual when a Formula leaf was available.

3. **Energy / Landauer.** Encoding, decoding, denoising, and gradient edits on latents erase and rewrite information. Landauer prices irreversible erasure as a thermodynamic lower bound and appears on receipts as a **labeled estimate**. Catalog or OpCounter estimates for latent paths remain **Estimated / Unmetered** on the soft-ref path (`measured_j=None`; `board_synth_claimed=false`).
4. **Certify-before-commit.** A low-energy latent state is a proposal until a certificate, settle, or typed refuse closes the act. Soft-ref close still binds `propose` → `certify` → `commit` / `refuse` → receipt. Latent score ≠ irreversible commit.

Teach-first takeaway. Latent depth is a compression technology. Mixture of Limits navigates whether that compression is cheapest-sufficient for the task, or whether refuse is the lawful success. See also §9: the question is not which latent generator wins a horse race; it is whether a generator should run at all.

8.5 Logical Intelligence: energy-based reasoning with LLMs and latents

Logical Intelligence (logicalintelligence.com) is a company building energy-based reasoning systems for constraint-heavy and mission-critical settings. Public materials (company blog and product pages, January 2026) teach a hybrid stack rather than a single chatbot:

- **Kona** is their core **energy-based model (EBM)** for reasoning (sometimes called an energy-based reasoning model, EBRM). An EBM assigns a scalar **energy** to a candidate state: low energy means more consistent with constraints and objectives; high energy means something is broken. Per their technical blog (Bodnia and Hanin, 21 Jan 2026), Kona is non-autoregressive at the *trace* level, globally scored over partial and complete traces, and reasons in a **continuous latent space** with dense vector tokens so local gradient-style edits can reduce constraint violations without regenerating an entire discrete prefix.
- **Aleph** is their orchestration / agentic layer that coordinates Kona, LLMs, and other tools. Public positioning: LLMs handle natural-language interface and candidate generation; the EBM layer evaluates and repairs under constraints. Aleph is also described as delivering verified / formal-reasoning workflows today (benchmark claims such as PutnamBench appear on their site; treat score numbers as **author-reported** unless independently reproduced here).
- Product framing on the Kona page: Kona is not marketed as a chatbot. Language models express and explore; Kona is positioned to evaluate what is valid or permissible before irreversible action in high-stakes domains (company examples include energy, manufacturing, semiconductor verification, robotics). Kona 1.0 was announced for partner pilots (company and press materials, Jan 2026). Leadership publicly listed includes founder and CEO Eve Bodnia and Yann LeCun as founding chair of a technical research board (company site / press).

Classical EBM backdrop (settled ML architecture). Energy-based models treat inference as finding low-energy configurations under a learned energy function. That framing is older than any one product; LeCun and others have long argued that reasoning can be cast as optimization over an energy landscape. Logical Intelligence's public thesis is that discrete, locally scored, autoregressive LLM traces scale poorly for long-horizon constraint satisfaction, and that continuous, globally scored EBM traces address that gap when paired with LLMs for interface.

Mixture of Limits reading (cascade map, not a product endorsement).

Logical Intelligence public piece	Cascade rhyme	Floor that still binds
Continuous latent trace + energy score	Soft rhyme with Solver / settle : optimize / repair under constraints	settle_refuse, Vol, energy estimate; low energy $\neq$ board joules
LLM for language / candidates	Model residual for expression and proposal	Model LAST; ModelGenerated $\rightarrow$ / Deterministic
Aleph orchestration among tools	Cheapest-sufficient routing among gears	Grammar coverage first; refuse when escalation is Vol-negative or unsafe
Constraint / proof-oriented close	Certify-before-commit spirit	Certificate / typed refuse before irreversible act

Verified here: the public architecture story and the cascade map—energy-based settle can rhyme with Solver; LLM residual stays Model LAST; latents do not erase Vol or Landauer floors. Author-reported latency and Sudoku figures stay field results (not soft-ref measured_j). Soft-ref path: board_synth_claimed=false. Unverified internal training details, unpublished energy-function forms, and AGI-completion claims remain **TBD** / out of scope.

8.6 World Labs: spatial intelligence and world models

World Labs (worldlabs.ai) is a frontier research and product company focused on **spatial intelligence**: models that perceive, generate, reason about, and interact with virtual and physical worlds across space and time. Co-founders publicly include Fei-Fei Li, Justin Johnson, Ben Mildenhall, and Christoph Lassner (company About page).

Teach the products and architecture first (public sources).

- **Marble** is their first product: generative 3D world models that create spatially coherent, persistent 3D worlds from images, video, text, and 3D layouts (company About).
- **Atlas** (company blog, 1 Sep 2026) is described as an omni **world model** pretrained to operate on text, images, video, and 3D. Architecture: a **multimodal autoregressive diffusion transformer**. Inputs are grounded in 3D to form a shared **spatial context**; the model generates what comes next while aiming for 3D consistency with what it has seen, and imagining what lies beyond. Public capability claims include camera-controlled video generation, sparse-view spatial reconstruction (including explicit 3D such as point clouds / Gaussian splats), space-time simulation for Real-to-Sim robotics workflows, and image / 360 generation. Atlas is positioned to power future Marble versions and is in early access with select partners. Author-reported benchmarks on camera-conditioned generation and 3D reconstruction appear on the Atlas post; cite the architecture and task list here, not those numbers as OpenIE measurements.

Fei-Fei's public spatial-intelligence framing (including the "from words to worlds" thesis) states that world models must handle semantic, physical, geometric, and dynamic complexity beyond today's LLM text corridor. That is a generator-class ambition: better world generators and simulators.

Mixture of Limits reading: even projected-superior world models hit floors.

1. **Grammar coverage.** Faithful reconstruction from enough views is not the same as a closed predictive law for a task. When a Formula or Lookup already answers the coordinate, generating a world is excess. Imagination that fills unseen regions is generative residual, not Deterministic commit.
2. **Certify-before-commit.** A spatially consistent video or splat is a proposal about geometry and appearance. Irreversible acts (robot motion, financial or safety-critical side effects) still require certify → commit/refuse → receipt. Soft-ref ReplayClass typing still applies: generative world output does not launder into Deterministic without a certificate path.
3. **Vol and energy.** Longer videos, denser 3D contexts, more denoising steps, and Real-to-Sim rollouts spend bits and joules. Vol stops when extra world detail does not change the decision benefit. Landauer bounds irreversible erasure as a labeled estimate; Atlas / Marble training or inference joules are field / author-reported, not soft-ref measured_j. Soft-ref path: board_synth_claimed=false.
4. **Materials embodiment.** High-fidelity world models intensify the applied-math × materials bottleneck: sensors, fabrics, and meters that make cheapest-sufficient close real at the edge, without laundering estimates as board watts.

World Labs systems optimize **which world generator / simulator runs** and how well it tracks space. Mixture of Limits asks whether that generator should run for the stated benefit, and which named floor closes first. Cross-link: §9 Mixture of Limits is not MoE table (outside-generation floors vs inside-generator capacity).

8.7 Thesis addendum: even methods projected to be superior still have limits

Latent compression, EBM settle hybrids (Logical Intelligence), and spatial world models (World Labs) can each be projected as superior to naive next-token MoE scaling for their stated jobs. Superiority inside a generative or hybrid corridor does not cancel Shannon pricing of bits, Howard pricing of Vol, Landauer pricing of irreversible erasure, or the require-to-certify before irreversible commit. Mixture of Limits is the **navigation law** that names those floors and binds Lookup → Formula → Solver/settle → Model LAST with refuse as lawful success. It is not a horse race among generators, latent depths, or world-model brands.

Soft-ref mo1 prove remains constructive existence **in software**. Soft-ref paths for Logical Intelligence, World Labs, or OpenIE do not stamp board package measured_j.

8.8 Test-time inference / test-time compute: more thinking is not free of floors

Teach first. Test-time inference (also called **test-time compute**) means spending additional computation *after* training, when answering a query: longer **chain-of-thought** (CoT) traces, search over candidate solutions, **majority vote** across samples, verifier-guided selection, or **process**

reward scoring of intermediate steps. OpenAI's o1 / o3-style systems popularized sequential scaling of reasoning traces at inference (OpenAI, "Learning to reason with LLMs," 2024). Related process supervision trains reward models on step-level labels rather than only final answers (Lightman et al., "Let's Verify Step by Step," arXiv:2305.20050; ICLR 2024). Parallel strategies sample many short traces and aggregate; sequential strategies extend one long trace. Surveys of inference-time scaling for complex tasks document both families and their dependence on verifier quality (e.g. arXiv:2504.00294).

The industry narrative treats extra thinking tokens as a new scaling axis parallel to training tokens and parameters. That axis is real as *engineering*: longer verified search can raise accuracy on hard problems when the verifier is good and the budget is spent on the right difficulty band. Empirical work on o1-like models also shows diminishing returns, overthinking, and cases where longer CoTs degrade accuracy or where parallel majority-style methods scale better than unbounded sequential length (e.g. ACL 2025 findings on o1-like test-time scaling; related overthinking analyses).

Mixture of Limits reading. Extra inference tokens are still bits under a budget.

1. **Value of Information (Vol).** Another thinking step is another observation about an internal search state. When the marginal step does not change the decision benefit (B), Vol says stop. Majority vote that re-samples without changing (B) fails the same test.
2. **Grammar coverage.** If Lookup, Formula, or Solver already covers the coordinate, opening a long CoT generator is waste. Covered grammar closes before Model.
3. **Energy / Landauer.** Every additional sampled token and every verifier forward pass erases and rewrites information. Landauer prices irreversible erasure as a thermodynamic lower bound and appears on receipts as a **labeled estimate**. Catalog or OpCounter estimates for test-time paths remain **Estimated / Unmetered** on the soft-ref path (measured_j=None; board_synth_claimed=false).
4. **Certify-before-commit.** A process-reward score or majority winner is a proposal until certify → commitrefuse → receipt. Soft-ref ReplayClass typing still applies: a long reasoning trace does not launder into Deterministic without a certificate path.

Teach-first takeaway. Test-time compute is a budget allocation inside the generative corridor. Mixture of Limits navigates whether that budget should open at all. More thinking steps are not free of Vol, grammar, energy, or certify floors.

8.9 Super learning: stacking ensembles still hit floors

Teach first (academic term). **Super learning** (also called the **super learner**) is a cross-validated ensemble / stacking method from targeted learning: fit a library of candidate algorithms, collect out-of-fold predictions, then learn a meta-learner that combines them to minimize cross-validated risk (van der Laan, Polley, and Hubbard, 2007; Polley and van der Laan, "Super Learner In Prediction," U.C. Berkeley Biostatistics Working Paper 266). It is related to stacking as introduced by Wolpert (1992) and adapted by Breiman (1996). A discrete super learner selects the single best candidate by

cross-validated risk; an ensemble super learner learns weights (often non-negative and summing to one) over candidates. The oracle results for cross-validation selectors underwrite asymptotic optimality *relative to the library*, not freedom from information or energy floors.

Label the industry narrative separately. Public “superintelligence” scaling talk (systems that broadly outperform humans across domains) is a *product and aspiration narrative*, not the van der Laan estimator. Academic **super learning** / **super learner** names the cross-validated stack; it is not a product brand and is not marketing AGI.

Mixture of Limits reading. Ensembles and stacks are still Model-class residual when they generate; when they combine deterministic candidates they still face Vol and certify limits.

1. **Vol.** Another base learner or another fold is another observation. When the meta-learner cannot change (B), stop.
2. **Grammar.** If Formula or Lookup covers the coordinate, stacking neural candidates is excess enumeration under a different name.
3. **Energy labels.** Training and evaluating a library multiplies forward passes. Estimates remain estimates; soft-ref ensembles keep measured_j=None; board_synth_claimed=false.
4. **Certify.** A stacked prediction is still a proposal until a certificate or typed refuse closes the act. Oracle optimality inside a library does not coerce ModelGenerated to Deterministic.

Teach-first takeaway. Super learning is cross-validated ensemble selection under cross-validation. Mixture of Limits maps it onto floors: better combination of residual leaves does not retire Vol, grammar, energy, or certify.

8.10 Survey of live approaches treated as “the answer” (2025–2026)

Each row teaches the technique once with a verified cite, then one Mixture of Limits floor sentence. This is not an uncited acronym dump. Author-reported speedups and benchmark scores are **field results**, distinct from OpenIE soft-ref measurements.

Approach	What it is (teach once)	Verified cite	Floor that still binds
Speculative decoding	A small drafter proposes several tokens; the large target model verifies them in parallel, preserving the target distribution while cutting serial decode steps	Leviathan, Kalman, and Matias, “Fast Inference from Transformers via Speculative Decoding,” ICML 2023 (PMLR); see also §9 HCSpec / CAS-Spec field rows	Speeds generation; does not refuse generation when Vol or grammar says stop
Mixture of experts (MoE)	Sparse gating routes each token to a few expert feed-forward modules so total parameters grow faster than active compute	Shazeer et al., “Outrageously Large Neural Networks,” ICLR 2017 (arXiv:1701.06538); Lepikhin et al., GShard, 2020 (arXiv:2006.16668);	Capacity <i>inside</i> the generative corridor; Mixture of Limits floors live <i>outside</i> generation (§2, §9)

Approach	What it is (teach once)	Verified cite	Floor that still binds
		Fedus et al., Switch Transformers, JMLR 2022	
State-space models (SSMs)	Sequence models with a compact recurrent state and near-linear scaling in context (selective SSM / Mamba) as an alternative or complement to attention	Gu and Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces,” arXiv:2312.00752 (2023); Gu, Goel, and Ré, S4, ICLR 2022	Efficient sequence leaf; still Model LAST when used as generator; Vol and certify unchanged
Retrieval-augmented generation (RAG)	Retrieve documents from an external index, then condition the generator on those passages so knowledge need not live only in weights	Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS 2020 (arXiv:2005.11401)	Retrieval can rhyme with Lookup when the hit is certified; uncertified retrieved text remains proposal, not Deterministic commit
Reasoning models / CoT scaling	Models trained or prompted to emit long intermediate reasoning (o1/o3-style, DeepSeek-R1, QwQ, and cousins)	OpenAI o1 public writeup (2024); DeepSeek-AI et al., DeepSeek-R1 technical report (2025); §8.8	Extra reasoning tokens still hit Vol and energy floors; longer is not always better
Diffusion language models	Discrete or continuous diffusion denoising over tokens or latents rather than purely left-to-right next-token prediction (e.g. score-entropy discrete diffusion)	Lou, Meng, and Ermon, “Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution” (SEDD), ICML 2024 (PMLR v235)	Denoising steps spend bits and joules; Landauer remains a labeled estimate only; certify before commit
Linear attention / retention variants	Attention alternatives with linear or chunkwise cost for long context (e.g. Retentive Network / RetNet multi-scale retention)	Sun et al., “Retentive Network: A Successor to Transformer for Large Language Models,” arXiv:2307.08621 (2023)	Cheaper long context is algorithmic efficiency; floors still stop when Vol or grammar is covered

Teach-first takeaway. Every live “answer” above optimizes *how* a generator runs or *which* parameters activate. Mixture of Limits asks whether the generator should run for the stated benefit, and which named floor closes first.

8.11 AI21 Jamba, then DeepSeek / GLM: architecture brand vs implementation efficiency

Teach the public Jamba story first. **Mamba** is a selective state-space sequence model with linear-time scaling in context (Gu and Dao, 2023). **AI21 Labs** positioned **Jamba** as a production-grade hybrid: interleaved Transformer attention layers and Mamba layers, plus mixture-of-experts on some MLPs, aimed at high throughput and a smaller key-value cache on long contexts (Lieber et al.,

“Jamba: A Hybrid Transformer-Mamba Language Model,” arXiv:2403.19887; AI21 blog announcing Jamba; follow-on Jamba-1.5 at larger active/total parameter counts, arXiv:2408.12570). Public claims emphasized hybrid efficiency (throughput and memory vs pure Transformers of similar class) and long context (up to 256K tokens in the released configurations). Author-reported benchmark and throughput numbers are **field results**, not OpenIE soft-ref measured_j.

What moved the frontier next was not a single architecture brand. Later open releases from Chinese labs showed that **algorithmic implementation**, training recipe, and inference engineering (often still on Transformer + MoE backbones) could take market share without waiting for the newest Western accelerator stack as the only path:

- **DeepSeek.** DeepSeekMoE (Dai et al., ACL 2024) and DeepSeek-V3 (DeepSeek-AI, arXiv:2412.19437) combine fine-grained MoE, multi-head latent attention, and systems co-design (including reported FP8 mixed-precision training and pipeline overlap). DeepSeek-R1 applies large-scale reinforcement learning for reasoning traces on that efficient base. Public technical reports emphasize cost-effective training relative to closed peers; treat GPU-hour and benchmark figures as **author-reported**, not soft-ref meters.
- **GLM / Zhipu.** The GLM family (Zhipu AI / Z.ai) progressed from dense GLM-4-class models to MoE reasoning releases such as GLM-4.5 (arXiv:2508.06471), with large total parameter counts and smaller active counts per token, plus agentic / reasoning post-training. Again, scores and active-parameter claims are field/author-reported.

Other Chinese and open labs followed similar patterns: MoE routing, quantization, speculative decoding, and training-systems efficiency on hardware that Western narratives sometimes treated as second-tier. The precise competitive ranking among vendors is out of scope and changes by week. What is in scope for Mixture of Limits is the structural lesson.

Mixture of Limits reading.

1. Jamba-class hybrids are a legitimate efficiency leaf: SSM + attention + MoE can cut memory and raise throughput for long generative contexts. They remain generators. Floors still bind.
2. DeepSeek / GLM-class results show that **implementation efficiency** (algorithms, precision, routing, RL post-training) moves the cost frontier. That is applied engineering, not a repeal of Shannon, Howard, or Landauer.
3. No architecture announcement retires Vol, grammar coverage, labeled energy estimates, or certify-before-commit. Soft-ref path: `board_synth_claimed=false`; Jamba / DeepSeek / GLM joules remain field / author-reported.

Teach-first takeaway. Architecture fads and lab efficiency wins are real. Mixture of Limits states why they matter and where they stop: cheaper generation still stops when Vol, grammar, or energy says stop.

8.12 Hardware economics thesis: computers are hardware; floors outlast fads

State as the study's claim (David Charlot / OpenIE framing), grounded in the sections above.

1. **Computers are hardware.** Every program runs on materials and devices: gates, memory, interconnects, power delivery, cooling. There is no software that escapes physics.
2. **Software is written for hardware.** Instruction sets, memory hierarchies, batch sizes, and quantization schemes exist because of devices. Software is **human applied engineering** under constraints and under economic / survival motivation. It is not a separate rocket-science realm detached from cost, yield, and joules.
3. **China showed the West a margin.** Inefficient deployment on the newest hardware leaves room. Optimized algorithms, MoE routing, training-systems co-design, and inference tricks on older or cheaper accelerators can take market share. The DeepSeek / GLM public efficiency stories (§8.11) are instances of that pattern; so are speculative decoding and quantization stacks (§8.10). Treat vendor scoreboards as field reports, not OpenIE meters.
4. **Mixture of Limits reading.** Architecture fads (Jamba hybrids, mixture of experts, test-time compute, diffusion language models, linear attention, retrieval-augmented generation, world models, energy-based hybrids) do not retire floors. The bottleneck remains **applied mathematics embodied in materials and devices**, plus labeled joule accounting. Algorithmic efficiency is real and is exactly why floors matter: cheaper implementations still stop when Vol, grammar, or energy says stop. Estimates (estimated_j, Landauer as labeled estimate) are not board package energy. Soft-ref path: measured_j=None; board_synth_claimed=false.

Bridge. §8.0.1 already stated that awareness of floors is not the gap; embodiment is. §8.12 adds the economic corollary: markets reward implementations that extract more work per device-dollar, and Mixture of Limits is the navigation law that keeps those implementations honest about when more work buys nothing.

9. Mixture of Limits is not MoE, and the model-to-model SOTA table

9.0 Spine (history + mathematics, not a horse race)

Mixture of Limits is proven as an **information-theoretic** approach rooted in **physics**: formula/law discovery in the Newton-Kepler lineage, not excess generation. Academia already has Shannon → Landauer → complexity/Vol → symbolic and energy floors; Mixture of Limits **embodies** those floors in a commit/refuse cascade. The remaining bottleneck is applied math × materials (hardware), not an awareness gap. The table below situates field systems so readers see what they optimize (**which neural generator runs**) without turning Mixture of Limits into another SOTA horse race. For latent depth, Logical Intelligence EBM hybrids, World Labs world models, test-time compute, super learning, live approach survey, Jamba/DeepSeek/GLM efficiency, and hardware economics, see §8.4-§8.12: even projected-superior methods still hit the same floors. Mixture of Limits asks whether a generator should run at all, converging to Vol/floors the way physics converged to predictive laws.

9.1 Law contrast (not a rebrand)

Mixture of Experts (MoE) places experts **inside** a generative corridor: a gate selects parameters; success is next-token likelihood; capacity grows by adding experts to the same act. Mixture of Limits places named **floors outside** generation: Lookup → Formula → Solver/settle → **Model LAST**; success is closed grammar + certified commit/refuse + joule receipt; the model is a demoted residual leaf, cold by default (allow_model=false).

Axis	MoE / model?model field	Mixture of Limits
Where the mixture lives	Inside the model (experts, draft/target pairs, strong/weak LLMs)	Outside generation (named floors + cascade gears)
Routing object	Which model (or draft head) answers	Which tier closes: Lookup / Formula / Solver / Model
Refuse	Rare; usually escalate to a larger model	First-class: Vol, grammar, settle, certificate, primitive_gap, fabric_unavailable, ...
Energy labels	Often latency/cost proxies; some papers meter GPU energy for LLM serving	Receipts: estimated_j labeled; soft-ref measured_j=None; board_synth_claimed=false
Success metric	Tokens/s, cost@quality, speculative speedup	Certified commit + typed ReplayClass + receipt

RouteLLM, GreenServ, and HCSpec optimize **which neural generator runs** (cost, Wh, or decode speedup inside the generative corridor). Mixture of Limits asks whether a generator should run at all.

9.2 Field SOTA (all model?model routing / speculation / constrained generation)

Every row below is **model?model**: routers choose among LLMs; speculative methods draft-then-verify with draft/target models; constrained decoding still samples from an LLM under a grammar. None implements Lookup→Formula→Solver→Model LAST with refuse-to-commit as the close law.

Measurement labels: **Metered** | **Estimated** | **Unmetered**; use a paper's own method section for its label. Field joules stay field results. Soft-ref prove path: **Unmetered** (software-ref); board_synth_claimed=false.

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs Mixture of Limits
RouteLLM	<i>RouteLLM: Learning to Route LLMs with Preference</i>	arXiv: 2406.18665 · ICLR 2025	Preference router (strong?weak LLM)	Query → strong or weak LLM	Cost / quality (API \$); not Mixture of Limits	Model?model; no Lookup/Formula floor; no typed

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs Mixture of Limits
	<i>Data</i> (arXiv); ICLR 2025 venue title uses <i>from Preference Data</i>				Landauer receipts	refuse-to-commit
PEARL (routing)	<i>PEARL: Performance and energy aware routing for LLMs</i>	DOI: 10.1016/j.future.2025.108218 · FGCS 176:108218	Energy-aware multi-LLM router	Query → LLM under energy cap (EMM predicts energy)	Authors report energy-aware routing (GPU/infra); not imported as Mixture of Limits measured_j	Still picks a model; Mixture of Limits may refuse before any model
GreenServ	<i>GreenServ: Energy-Efficient Context-Aware Dynamic Routing for Multi-Model LLM Inference</i>	arXiv: 2601.17551 · ICPE 2026	Contextual bandit multi-LLM router	Query features → LLM; accuracy vs measured GPU energy (Zeus)	Metered (authors' GPU Wh via Zeus) on their LLM pool (field result, not Mixture of Limits board synth)	Model pool routing; Mixture of Limits cascade is non-neural first
VoltanaLLM	<i>VoltanaLLM: Energy-Efficient and SLO-Aware Disaggregated LLM Serving via Adaptive Frequency Control and State-Space Routing</i>	arXiv: 2509.04827	P/D-disagg serving + frequency + state-space route	Prefill/decode instances + GPU frequency under TTFT/ITL SLOs	Metered (authors; pyNVML on A100/GH200) up to ~36.3% E2E GPU energy vs max-freq baseline (field serving result)	Routes instances of the same generative stack ; Mixture of Limits may never open decode
HCSpec	<i>HCSpec: Two-Tier Horizontal</i>			Position-specialized	Latency speedup (vs	Speeds generation;

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs Mixture of Limits
	<i>Cascade Speculative Decoding for High-Efficiency Large Language Model Inference</i>	DOI: 10.18653/v1/2026.acl-long.353 · ACL 2026	Speculative decoding (draft [?] target)	draft cascade → target verify	EAGLE-3 / AR); energy not Mixture of Limits claim	does not refuse generation on floors
CAS-Spec	<i>CAS-Spec: Cascade Adaptive Self-Speculative Decoding for On-the-Fly Lossless Inference Acceleration of LLMs</i>	arXiv: 2510.26843 · NeurIPS 2025	Self-speculative cascade (DSIA drafts)	Draft stages from target (sparsity/quant) + DyTC	Latency speedup ~1.1x-2.3x AR (authors); lossless tokens	Model-internal draft hierarchy ≠ Mixture of Limits floors
GCD	<i>Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning</i>	arXiv: 2305.13971 · DOI: 10.18653/v1/2023.emnlp-main.674 · EMNLP 2023	Grammar-constrained LLM decoding	Mask logits to CFG	Unmetered / quality metrics	Constrains tokens ; Mixture of Limits constrains commit
CRANE	<i>CRANE: Reasoning with constrained LLM generation</i>	arXiv: 2502.09061 · ICML 2025 (PMLR v267)	Reasoning-augmented constrained decoding	Alternate unconstrained reason [?] constrained answer	Accuracy on GSM-symbolic / FOLIO; not Mixture of Limits joules	Still LLM generation under grammar; Mixture of Limits can close at Formula without an LLM

Name collision (do not conflate): ICLR 2025 also has *PEARL: Parallel Speculative Decoding with Adaptive Draft Length* (arXiv:[2408.11850](#)); speculative, not the FGCS energy router. §9's PEARL row is the **routing** paper (DOI 10.1016/j.future.2025.108218). Speculative PEARL sits in the same model[?]model bucket as HCSpec/CAS-Spec if cited later.

9.3 What Mixture of Limits claims instead

From PLAN prove (see §6):

- 1. **Cascade order is law**, not a heuristic over LLMs: Lookup → Formula → Solver/settle → Model LAST.
- 2. **Refuse-to-commit** is typed and receipted (voi, settle_refuse, efa_certificate, primitive_gap, fabric_unavailable, claim_unknown, ...).
- 3. **Receipt labels** on the proven path: measured_j=None (Unmetered), estimates labeled (**Estimated**), board_synth_claimed=false.
- 4. **ReplayClass** cannot strengthen: ModelGenerated →/Deterministic.
- 5. Soft-ref multi-fabric routing chooses DeviceKind after tier close; still not “pick GPT-4 vs Mixtral.”

9.4 Measurement bounds

Soft-ref / paper-results bound	Allowed
Soft-ref prove path measured_j	Soft-ref prove with measured_j=None (Metered only when a probe returns)
Board synth flag without meter + synth log	board_synth_claimed=false
GreenServ / VoltanaLLM Wh as soft-ref board energy	Cite them as field model-to-model SOTA with their own Metered labels
Tokens/J horse race vs HCSpec	Refuse-before-tokens as the navigation win

Field SOTA above remains **model?model**. Mixture of Limits contribution is not a better router among generators: it is the navigation law that embodies known information/energy floors, with a prove harness that keeps Model last and refuse first-class.

10. Toward AGI via limits: open questions and companions

If AGI-grade reliability means closing tasks with certificates, receipts, and known floors, not maximizing tokens under an unlimited generative prior, then the historical method already points the way. Kepler and Newton pursued limits: the shortest predictive law that covered the grammar. Shannon priced bits; Howard priced Vol; Landauer and Bennett priced irreversible erasure; Kolmogorov, Solomonoff, and Chaitin priced compression and incompleteness. Recent formula/mechanism systems (AI Feynman, SINDy, PySR, AI-Newton, AlphaGeometry, MWMs) keep proving the same pattern in modern form.

The missing piece is not the idea. Shannon, Howard, Landauer, Bennett, Kolmogorov, Solomonoff, and Chaitin state the floors as settled theory. The bottleneck Mixture of Limits asserts is **applied mathematics embodied in materials and physical hardware**: devices, fabrics, and meters that make cheapest-sufficient close real at the edge, without laundering estimates as board watts, and without treating MoE capacity as a substitute for grammar coverage.

Mixture of Limits says CI should navigate those floors explicitly: Lookup and Formula first, Solver when settle is the right physics, Model last, refuse when Vol or safety says stop. Soft-ref prove shows the navigation law is constructive in software. Companions on this site (NI commit record; Satiation economic done) supply interfaces and economic stops. Soft-ref path keeps `board_synth_claimed=false` and `measured_j=None` unless Metered.

Open questions

- Full Periodic Stack 258 live catalog and calibrated () corpora (subset navigator is in soft-ref proof; full catalog is not).
- Live WCA MCP / System One pre-gate and leapfrog openie-path ask (adapters remain stubs on the proven path).
- Materials / device paths that make Formula+LUT coin-cell closes dominate generative spend, with Metered probes when package joules are claimed.
- When a Metered `measured_j` path is published beside soft-ref estimates without laundering estimate as board watts.
- How Vol thresholds should be set per domain without smuggling engagement metrics as completeness (see Satiation).
- Whether mechanism libraries (MWM cousins; AI-Newton concept base) should be imported as typed Lookup entries without breaking clean-room prove.
- How to bind third-party EBM settle scores (e.g. Logical Intelligence-style energies) and world-model proposals (e.g. World Labs Atlas/Marble) into certify-before-commit without laundering generative output as Deterministic.
- How to price test-time compute budgets under Vol so adaptive thinking stops when marginal CoT tokens do not change (B), without treating longer traces as automatic progress.
- Whether academic super-learner libraries should be imported as typed Solver/ensemble leaves under certify, without equating them to industry superintelligence narratives.
- How to document algorithmic-efficiency wins (DeepSeek / GLM-class MoE and systems co-design; speculative decoding) as Estimated field results beside soft-ref receipts, keeping author GPU-hours distinct from Mixture of Limits `measured_j`.

Companions on this site

- [Notational Intelligence as Commit Law](#); commit/refuse record; analytical energy; measurement bounds.
- [Satiation and Scarcity after Free AI](#); economic done after free digital inference.
- Living figures: [/living/mol/](#) (companions; analytical OpCounter $\neq$ board power; package `measured_j` only when Metered).

Closing sentence. Pursuit of limits (compression into predictive law, priced information, thermodynamic accounting) is the through-line from Kepler and Newton to Shannon, Howard, Landauer, and Kolmogorov. Mixture of Limits is that through-line stated as a navigation law for computer intelligence; embodying it in materials remains the hard problem. Latents, EBM hybrids, world models, test-time compute, ensembles, and architecture-efficiency wins (§8.4-§8.12) refine generators and lower device-dollar cost; they do not retire the floors. Computers are hardware; software is applied engineering under constraints; labeled joule accounting binds.

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Kepler's laws / Newton's *Principia* / Brahe's observational program are treated as historical common knowledge in §8.1; imprint DOIs for critical editions **TBD** if a journal version requires them. Raiffa decision-analysis imprint DOI **TBD**. Press-wire details for Logical Intelligence beyond the company blog (exact pilot lists, internal AGI claims) are company-reported. Author-reported GPU-hours, throughput, and benchmark scores for Jamba, DeepSeek, GLM, speculative decoding, and test-time scaling papers are **field / author-reported**, not soft-ref measured_j.

Appendix A. Measurement bounds

- Soft-ref prove path: board_synth_claimed=false; measured_j=None unless a Metered probe returns
- Landauer = labeled thermodynamic lower-bound estimate on receipts
- Analytical / catalog estimated_j ≠ board power
- Soft-ref mol prove ~29 VERIFIED = constructive existence in **software**
- Clean-room Mixture of Limits: law+runtime, not an MoE wrapper
- §§6 and 9 Leapfrog placeholders intentionally incomplete
- DOIs only when verified via search/source inventory; otherwise marked TBD
- Bottleneck: materials/applied-math embodiment. Settled IT/physics already name the floors
- §8.4-§8.12 map latents / Logical Intelligence / World Labs / test-time compute / super learning / live approaches / Jamba-DeepSeek-GLM / hardware economics onto floors—not a generator horse race; field product joules stay field; estimates ≠ board package energy

Appendix B. Reproducibility pointers

```
# On a machine with the mixture-of-limits workspace:
cargo test --workspace
cargo run -p mol-cli -- prove # expect VERIFIED lines, exit 0
```

Site preview for this study (research-openie-web):

```
cd /Users/dcharlot/data-share/vibe-coding/research-openie-web
pnpm install
pnpm dev
# open /papers/mol/ , /about/mol , /living/mol/
```

PDF regeneration (optional; needs pandoc + weasyprint): `./scripts/generate-pdfs.sh` after mol is added to the script's gen_one list.